

Eddy Current Pulsed Thermography based Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition

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Abstract: Defect detection in non-destructive testing (NDT) is fundamental to structural integrity and quality control. Despite the unique advantages of thermal image analysis, especially Eddy Current Pulsed Thermography (ECPT), such as fast detection and in-depth information, existing signal processing and sparse decomposition algorithms often struggle with poor contrast, low resolution, and ambiguity in extracting weak defect signals from noisy thermal sequences. These limitations arise from problems such as manual feature selection and the computational burden of processing the entire thermal sequence. In this paper, we propose a new approach to improve defect detection in ECPT data to address the challenges of insufficient defect information and computational efficiency. We propose a keyframe extraction (KFE) mechanism using entropy and density clustering to select keyframes from the original thermal sequences, which significantly reduces the processing load without affecting the defect information. We then implement a fast, accelerated CUR decomposition based on a low-rank, sparse decomposition (IRCUR) to efficiently isolate weak defect information from the noise in these selected keyframes. This CUR-based decomposition further accelerates the computation by approximating low-rank and sparse components using a smaller matrix. Experimental results on several samples with different types of defects show that the proposed method can extract weak defect information quickly and accurately, obtaining superior performance compared to recent state-of-the-art algorithms and saving computation time significantly.

Keywords: Eddy current pulsed thermography, key frame extraction, accelerated sparse decomposition, non-destructive testing.

I. Introduction:

Structural health monitoring of metals directly affects the reliability of a mechanical system. It is equally important in assessing quality control, remaining usable life, and early fault identification. Accurate detection of surface defects in metals is therefore essential, as surface defects are among the most commonly encountered flaws in metallic components in service. NDT is one of the widely adopted techniques for defect inspection due to its unique advantages [1]. Specifically, eddy current pulsed thermography (ECPT) [2] is a popular NDT technique whose origins lie in eddy current testing. This technique uses excitation coils with high frequency to propagate eddy currents on a specimen under test. An infrared camera is used to capture the propagating heat patterns on the metal surface as thermal images. When defects exist on the metal surface, they induce an irregular eddy current distribution distinct from the non-defective areas, and this irregularity can be observed in the temperature profile of the specimen according to Joule's law.

Building on this principle, inductive thermography has become a mature NDT method for detecting surface cracks in metals [3]. A short high-frequency pulse through an induction coil induces eddy currents and Joule heating inside the specimen under test. The induced eddy current decays exponentially below the surface within the skin depth δ , defined as $\delta = \frac{1}{\sqrt{\pi f \mu_0 \mu_r \sigma}}$, where f is the excitation frequency, $\mu_0 = 4\pi \times 10^{-7}$ Vs/Am is the permeability of free space, μ_r is the relative magnetic permeability of the material, and σ is its electrical conductivity [3]. For ferromagnetic carbon steel with μ_r approximately 600 at $f = 200$ kHz, the skin depth is approximately 0.03 mm, which is far smaller than the typical surface crack depth of 0.5 to 2 mm. As a result, eddy currents are confined to a very thin near-surface zone and heat selectively along the crack walls, producing a characteristic hot-line thermal signature at the defect. Surface cracks also disturb the eddy current flow and heat diffusion, producing characteristic thermal patterns in the infrared images. The technique

is non-contact, fast, and enables large-area inspection. It also supports quantitative characterization: hot-spots at crack tips indicate crack length, while phase contrast helps estimate crack depth [4]. Reliability of inductive thermography is well established through probability of detection (POD) studies: $a_{90/95}$ values of 0.60 mm for ferritic steel, 0.78 mm for titanium, and 1.50 mm for nickel-based superalloys have been reported for short surface cracks using phase images [5,6]. These studies confirm that the thermal contrast at crack faces is governed by disruption of eddy current flow, and that reliable detection requires both sufficient excitation energy and appropriate signal processing to separate the defect-induced thermal anomaly from the background.

Most recently, ECPT as a fast-growing NDT inspection methodology has found applications in power transmission technology, electronics manufacturing, physics-based experimental systems, mechanical systems, and the transportation industry, owing to its non-contact nature, large scan area, and good resolution [7-13]. The quality of ECPT captured images is directly affected by two important factors: thermal diffusion and emissivity. Emissivity is measured as the ratio of radiation intensity on the specimen with reference to a blackbody object at constant temperature [14]. Generally, the emissivity of smooth surfaces is quite low, making it challenging for the infrared camera to capture intrinsic temperature variations on the specimen. Furthermore, uneven and irregular emissivity causes measurement errors. In ECPT-based defect inspection, many researchers have proposed a number of solutions from both software and hardware enhancement perspectives, such as radiation law-based algorithms to compensate results [15-17], addition of high-emissivity-based layers to enhance heat conduction [18,19], and approaches involving painting the specimen surface [20]. In the present experimental study, all specimens are ferromagnetic carbon steel components that were either rust-free or spray-protected with a thin coating to stabilise surface emissivity. Under the heating conditions used here, emissivity variation had less impact on image quality than lateral thermal diffusion, which was the dominant source of background degradation across all five specimens.

Thermal diffusion corresponds to the transmission of heat from high to low temperatures and significantly affects the quality of captured ECPT images [21]. Thermal diffusion is generally divided into vertical and lateral diffusion. Vertical diffusion corresponds to heat propagation from the surface into the interlayer of the specimen, forming the basis for subsurface detection in ECPT [22]. For resolving defect depth and size, vertical diffusion characteristics such as skewness and kurtosis can be used [23]. Lateral diffusion corresponds to heat propagation parallel to the specimen surface and generally has adverse effects on ECPT image contrast [24,25]. In surface defect detection, lateral diffusion blurs the boundary between defective and sound areas through the lateral conduction of heat from high-temperature eddy current zones into the surrounding cooler material, which reduces the thermal gradient at defect edges and makes it difficult to distinguish the defect from the sound area.

To resolve this and improve image quality, several authors have proposed spatial-based algorithms [26]. Resendiz-Ochoa et al. [27] proposed a thresholding-based image segmentation algorithm, deployed on thermographic images with dislocation and rod/bearing damage defects. Ratsakou et al. [28] proposed an enhanced Canny edge detection algorithm for thermographic sequences. Xie et al. [29] developed a classification algorithm based on support vector machines, classifying ECPT images using a region-based strategy and simple spatial statistical features such as mean, peak values, and variance. One advantage of spatial-based algorithms is that they process data very fast. However, they tend to overlook abnormalities corresponding to defect information for two primary reasons: first, the change in temperature over time due to heat diffusion creates a varying temperature region around the defect area, making it difficult to establish a clear boundary between the defect and the diffusion zone; and second, the magnitude and gradient are prone to random noise, and enhancing these spatial features inadvertently amplifies the noise.

Despite the absence of a clear boundary between the defective and diffusion areas, the change in temperature over time between these two regions is quite different. Furthermore, temporal-based features such as frequency and phase are less prone to random noise. Accordingly, researchers have proposed a number of temporal algorithms. The Fast Fourier Transform (FFT) based algorithm [30] converts the time-domain temperature data into the Fourier domain for better defect enhancement. The thermal signal reconstruction

(TSR) algorithm [31] converts the thermal time-domain data into the logarithmic domain for better contrast of the ECPT data. Another approach is the extraction of statistical features from the thermal data to distinguish defects from non-defective regions [32]. Ding et al. [33] proposed the use of skewness features for corrosion detection in ECPT images. Tu et al. [34] proposed transient pattern-based temperature separation algorithms including Independent Component Analysis (ICA) and Principal Component Analysis (PCA) for detecting defects in composite insulators using ECPT images. Further, the combination of temporal and spatial features through matrix and tensor-based sparse and low-rank decompositions has become a major research focus for ECPT-based defect detection and enhancement.

In this regard, authors in [35] proposed an online stochastic tensor decomposition method for multispectral video frames for background and foreground modelling. In [36], authors proposed an improved low-rank and sparse decomposition model using PCA-based SVD with an improved tensor nuclear norm for background-foreground modelling in video sequences. In [37], authors proposed an algorithm for CP tensor decomposition using an alternating least squares method, employing both QR and SVD decomposition to find low-rank and sparse components in multidimensional data. In [38], authors proposed a novel low-rank and sparse decomposition method using the mixture of Gaussians (MoG) distribution for surface defect detection in metal components. In [39], authors proposed a novel low-rank and sparse decomposition approach using CUR decompositions instead of SVD. In [40], authors proposed an image entropy and density-based approach to extract keyframes from video sequences, tested on hand gesture recognition data with good results. In [41], authors presented an enhanced diffusion-based method (EDBM) for ECPT to combat thermal diffusion blurring by deriving temporal features from spatial features, achieving efficient in-line defect detection with significantly reduced computational cost and improved accuracy.

Temporal-based methods perform better than spatial-based methods because thermal diffusion is continuous throughout the whole ECPT image sequence. However, to find temporal features, the complete ECPT video sequence must be processed using complex matrix/tensor-based operations, which increases the overall computational cost. Both spatial and temporal approaches have their own trade-offs: spatial-based algorithms are computationally inexpensive but less accurate, while temporal-based algorithms are more accurate but slower. Achieving both high accuracy and fast processing is a challenging task, and existing methods present a trade-off. Deploying sparse and low-rank decomposition algorithms on the whole ECPT sequence, which contains noise information in most frames and weak defect information in only a few frames, makes the problem more difficult and computationally expensive. In this paper, we propose a key frame extraction mechanism with CUR decomposition based low-rank and sparse decomposition of the original thermal sequence from the ECPT experiments. The highlights of the proposed approach are given as;

- 1) Instead of using the raw thermal sequence for low-rank and sparse approximation we deploy an entropy and density clustering-based mechanism to select the few key frames from the original raw sequence, this in turn will lower the processing burden on the LR and sparse algorithm saving significant processing time while preserving the defect extraction.
- 2) Further we propose the fast and accelerated CUR decomposition based low-rank and sparse decomposition approach to extract the weak defects information from the noise in thermal data. By deploying the CUR based decomposition we can accelerate the computation time on the thermal data by using small CUR matrix to approximate low-rank and sparse components from the thermal data while preserving the defect information extraction.

The rest of the paper is organised as follows. Section II discusses the proposed methodology. Section III discusses the experimental setup and specimen information. Section IV discusses the experimental results and analysis. Section V presents the conclusions.

II. Proposed Methodology

In this paper, we propose the key frame extraction methodology along with the robust CUR decomposition based low-rank approximation algorithm. The key frame extraction helps remove unwanted and redundant information from the video sequence greatly and robust CUR based low-rank decomposition approach helps extract weak defects information from the thermographic sequences.

Let S_{kF} be the key frames to be extracted from the overall thermographic sequence X . The mathematical formulation is set as:

$$S_{kF} = F_{kF}(X) \quad (1)$$

where F_{kF} is the frame extraction approach. This key frame extraction approach utilizes the entropy and density clustering to achieve the results.

The Entropy for a given frame can be calculated as;

$$E(f_{ri}) = -\sum_j q_{f_{ri}}(j) \log q_{f_{ri}}(j) \quad (2)$$

where $q_{f_{ri}}(j)$ is the pdf of a frame f_{ri} . Next, we extract the local minimum and maximum points from the entropy data as;

$$P_{maximum} = \{E(f_{ri}), \text{ if } E(f_{ri}) > E(f_{ri+1}) \& E(f_{ri}) > E(f_{ri-1})\} \quad (3)$$

$$P_{minimum} = \{E(f_{ri}), \text{ if } E(f_{ri+1}) > E(f_{ri}) \& E(f_{ri-1}) > E(f_{ri})\} \quad (4)$$

Next, we use P_{ext} as the union of $P_{maximum}$ and $P_{minimum}$ which helps extract local representation within the whole frame sequence which in turn benefits the overall representation in the thermographic sequence.

Further these extreme points P_{ext} are clustered using the density clustering step algorithm of [40]. In density clustering step algorithm first finds the local maximum points in a cluster and then goes from higher to lower density sequentially. For each data point in P_{ext} we find the local density $D_{P_{ext}}$ and $\sigma_{P_{ext}}$ which is the distance from higher density points.

$$D_{P_{ext}} = \sum_{P_l} T(d_{P_{ext} P_l} - dc) \quad (5)$$

where $T(x) = 1$ if $x < 0$ and $T(x) = 0$ otherwise

dc is the cutoff and $d_{P_{ext} P_l}$ is the distance between data points. $\sigma_{P_{ext}}$ can be calculated as;

$$\sigma_{P_{ext}} = \min_{P_l : D_{P_l} > D_{P_{ext}}} (d_{P_{ext} P_l}) \quad (6)$$

This gives the minimum distance between points. Finally, S_{kF} are the N largest σ values of the corresponding P_{ext} from where N signifies the number of frames. Selecting the S_{kF} further we use the CUR decomposition based low-rank approximation to extract the defect information from the thermographic sequences. The key frame extraction approach helps us bring down the total no of frames from the original thermographic sequence by removing redundant frames and keeping only the small number of important frames. This approach also helps reduce the processing time for the CUR based low-rank approximation algorithm.

Let X be the S_{kF} extracted from the original thermographic sequence. Further we decompose X in low-rank components Q and *Sparse* component P and this problem can be formulated as;

$$\min_{P, Q} \|X - P - Q\|_F \quad \text{s.t.} \quad \text{rank}(Q) \leq r \text{ and } \|P\|_0 \leq \alpha n^2 \quad (7)$$

where r represents the rank, $\|\cdot\|_0$ represents the l_0 -norm, $\|\cdot\|_F$ is the Frobenius norm, α is the sparsity level and for simplicity $n_1 = n_2 = n$ is used to extend the result to the non-square matrices. The solution to eq: 7 is proposed by authors in [39] by using the CUR decomposition to iterative model replacing conventional SVD approach which in turn helps reduces the computational complexity of the overall solution. The overall pseudocode of the proposed algorithm is shown in (Algorithm-1).

Algorithm-1: Robust *KFE* based *CUR* for thermographic data

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1. Input: $X = S_{kF}$ observed thermographic keyframe matrix of dimension $(H \cdot W) \times 7$ (seven keyframes selected by the *KFE* step; $r = 4$ (target rank of the low-rank component Q); $\epsilon = 10^{-5}$ (convergence tolerance); $\Delta_0 = \max|X|$ (initial thresholding value, the maximum absolute entry of the input matrix); $\beta = 1$ (thresholding decay parameter); $|A|$ and $|B|$ (uniformly resampled row and column index sets at each iteration, following [39]).
 2. Sample row indices $|A|$ and column indices $|B|$ in a uniform manner.
 3. $Q_0 = 0, P_0 = 0, k = 0$
 4. while $e_k > \epsilon$: do $\therefore e_k$ will be defined below.
 5. $\Delta_{k+1} = \beta^k \Delta_0$
 6. $[P_{k+1}]_{:,B} = \mathcal{T}_{\Delta_{k+1}}[X - Q_k]_{:,B}$
 7. $[P_{k+1}]_{A,:} = \mathcal{T}_{\Delta_{k+1}}[X - Q_k]_{A,:}$
 8. $C_{k+1} = [X - P_{k+1}]_{:,B}$
 9. $U_{k+1} = H_r([X - P_{k+1}]_{A_i B})$
 10. $R_{k+1} = [X - P_{k+1}]_{A,:}$
 11. $Q_{k+1} = C_{k+1} \cdot U_{k+1}^+ \cdot R_{k+1}$
 12. $k = k + 1$
 13. end while
 14. Output : C_k, U_k, R_k : *CUR decomposition of Q , P : Sparse component*
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Normally for the low-rank component *SVD* algorithm is used which may incur higher cost assuming (n) is large. Here *CUR* decomposition is deployed to represent the low-rank component Q as;

$$C_{k+1} = [X - P_{k+1}]_{:,B} \quad (8)$$

$$R_{k+1} = [X - P_{k+1}]_{A,:} \quad (9)$$

$$U_{k+1} = H_r([X - P_{k+1}]_{A_i B}) \quad (10)$$

Where A and B are the uniform sampling indices (rows and columns) and $H_r(\cdot)$ is the best rank approximation, the updated low-rank component can now be given as;

$$Q_{k+1} = C_{k+1} \cdot U_{k+1}^+ \cdot R_{k+1} \quad (11)$$

Now for updating the sparse component hard thresholding methods are used as;

$$[\mathcal{T}_\Delta Y] = \begin{cases} Y_{i,j} & |Y_{i,j}| < \Delta \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

For each iteration with the selected thresholding value Δ_{k+1} , the sparse matrix P_{k+1} is obtained keeping $\|P - P_{k+1}\|_\infty$ in control. The value of Δ_{k+1} can be evaluated / selected as $\Delta_{k+1} \geq \|Q - Q_k\|_\infty$ which ensures that the $\text{supp}(P_{k+1}) \subseteq \text{supp}(P)$ and $\|P - P_{k+1}\|_\infty \leq 2\Delta_{k+1}$.

As in the update of low-rank component Q , we used indices A, B to sample the values from Q , therefore this approach helps avoid thresholding the entire data $X - Q_k$, and only that data is threshold provided by indices A, B , i.e. evaluating and forwarding of data $[P_{k+1}]_{A,:}$ and $[P_{k+1}]_{:,B}$ is only required this in turn requires the computation of only $[Q_k]_{A,:}$ and $[Q_k]_{:,B}$.

Now to update the sparse component P we evaluate

$$[Q_k]_{A,:} = [C_k]_{A,:} \cdot U_k^+ \cdot R_k \text{ and } [Q_k]_{:,B} = C_k \cdot U_k^+ \cdot [R_k]_{:,B} \text{ followed by the hard thresholding.}$$

The stopping criteria e_k is computed as;

$$e_k = \frac{\|[X - Q_k - P_k]_{A,:}\|_F + \|[X - Q_k - P_k]_{:,B}\|_F}{\|X_{A,:}\|_F + \|X_{:,B}\|_F} \quad (13)$$

III. Experimental Setup:

The ECPT detection system works in reflection mode by installing a shuttle shaped induction coil combined with an infrared camera system facing the surface of the specimen. The induction coil system is hooked up to a high-frequency power generator Easyheat-224, that supplies pulsed excitation current to the specimen and the water-cooling systems is used for the cooling phase. The system operates at an excitation frequency of 200 kHz with a peak excitation current of 200 A. Data acquisition spans a total observation window of 500 ms, consisting of a 200 ms heating pulse followed by a 300 ms cooling phase. The coil is set at a lift-off distance of between 5 mm from the specimen surface. If a principal crack direction was established, the coil was set perpendicular to the crack line so as to maximise disruption of the eddy currents at the crack surfaces. For specimens with irregular crack patterns or multiple crack directions, the coil was positioned to ensure equitable inspection coverage. Fig. 1 shows the schematic of the ECPT detection system and Fig. 2 shows the actual laboratory platform used in all experiments.

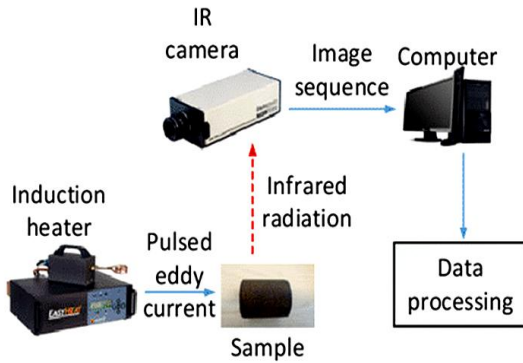


Figure 1: Schematic of the ECPT inductive thermography detection system

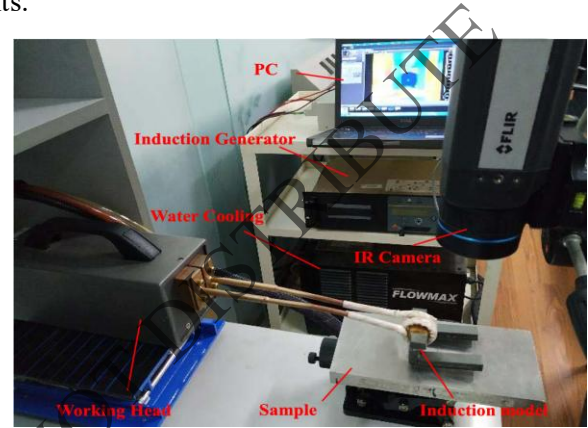


Figure 2: Photograph of the experimental platform

A FLIR A655sc uncooled infrared camera was used to record thermal images. The output images contain 640×120 pixels for 200 Hz or 640×240 for 100 Hz pixels, which for the working distances used translated to a ground resolution of about 0.15-0.20 mm/pixel, constant for all five specimens. The camera's thermal sensitivity (NETD) is below 50 mK. The entropy and density clustering mechanism described in Section II were used to select seven keyframes from the specimen. All processing was performed in MATLAB 2021b on a PC with Intel Core i5 processor, 16 GB of RAM, Windows 11, no GPU support.

This study utilized five ferromagnetic carbon steel specimens, with a naturally occurring defect on the surface. Table I shows the photographs of all five specimens with the crack locations marked by red rectangles. All cracks are natural defects from fatigue, welding, and service-induced stress. There were no artificial defects present in the samples and no machining or EDM cutting was used to create them. Other specimens had no rust, but some had been treated with a thin spray coating to stabilise emissivity. For a quantitative evaluation, a signal region S_{area} and a background noise region N_{area} of equal area were manually defined for each specimen before any algorithm evaluation, and both regions were fixed and applied identically to all seven compared methods. All cracks considered in this study are naturally occurring service-induced defects whose exact depth has not been quantified by any cross-sectional measurement. The detected cracks, based on the skin depth (approximately 0.03 mm at 200 kHz) and the thermal signatures, are all consistent with a superficial breakage in defect depth of about 0.5 to 2 mm, which is the typical detection range for inductive thermography at the specified excitation frequency [3,4]. Figure 3 represents thermal frames of the five evaluated specimens illustrating the Region of Interest (ROI) selection.

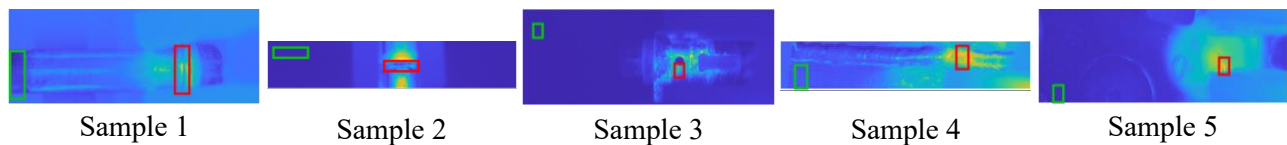
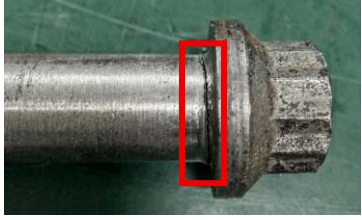
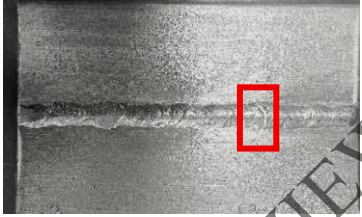


Figure 3: ROI Definitions - Red: Defect (S_{area}) and Green: Background (N_{area})

Table I: Specimen Photographs and Information

Type	Image	Material	Frame Size (W x H)	Defect Description
Sample 1 Threaded shaft (screw-type)		Ferromagnetic carbon steel	640 x 240	Natural fatigue crack at the root of the threaded section; crack is exceptionally narrow, irregular, and discontinuous; crack location close to the thread root edge, making detection difficult with standard configurations
Sample 2 Threaded shaft (screw-type)		Ferromagnetic carbon steel	640 x 120	Natural fatigue crack at the root of the threaded/splined section; thin and irregular crack morphology; complex surface geometry near the crack edge
Sample 3 Flanged component		Ferromagnetic carbon steel	640 x 240	Natural surface crack on a component with complex non-uniform geometry; irregular crack shape; non-uniform electromagnetic heating due to component geometry
Sample 4 Weld joint		Ferromagnetic carbon steel	640 x 120	Natural weld crack formed during welding; crack is too small to observe directly and has an irregular discontinuous shape; surface carries dense welding scales that generate intense structured thermal signals
Sample 5 Cast plate		Ferromagnetic carbon steel	640 x 240	Natural surface crack on a flat cast plate with complex surface features and holes; crack is irregular and located near a surface marking

IV. Experimental Analysis:

All results are presented as normalized sparse intensity maps using a consistent parula colormap. Performance is evaluated using SNR and AUROC. A defect region S_{area} is placed over the crack and a background region N_{area} on a nearby defect-free surface, with identical size and fixed placement across all methods [42]. The SNR is computed as:

$$SNR = 20 \log_{10} \left(\frac{|\mu_S - \mu_N|}{\sigma_N + \epsilon} \right) \quad (14)$$

where $\mu_S = \text{mean}(S_{area})$ and $\mu_N = \text{mean}(N_{area})$ are the mean pixel intensities of the sparse component in the defect and background regions, $\sigma_N = \text{std}(N_{area})$ is the corresponding standard deviation of the

background region, and ϵ is a small stability constant to avoid division errors. The numerator represents defect-to-background contrast, while the denominator penalizes high-variance responses.

AUROC evaluates pixel-level separability independent of a fixed threshold. Pixel values from S_{area} are treated as positive samples and from N_{area} as negative samples. A threshold sweep generates true positive rate (TPR) and false positive rate (FPR) pairs, whose trapezoidal area defines AUROC [43]. TPR and FPR are computed as:

$$TPR = \frac{TP}{N_{pos} + \epsilon}, FPR = \frac{FP}{N_{neg} + \epsilon} \quad (15)$$

where TP and FP are the numbers of correctly and incorrectly classified pixels, and N_{pos} and N_{neg} are the total numbers of defect and background pixels and ϵ again is a small stability constant to avoid division errors, respectively. The trapezoidal area under the resulting curve defines the AUROC [43], mathematically expressed as:

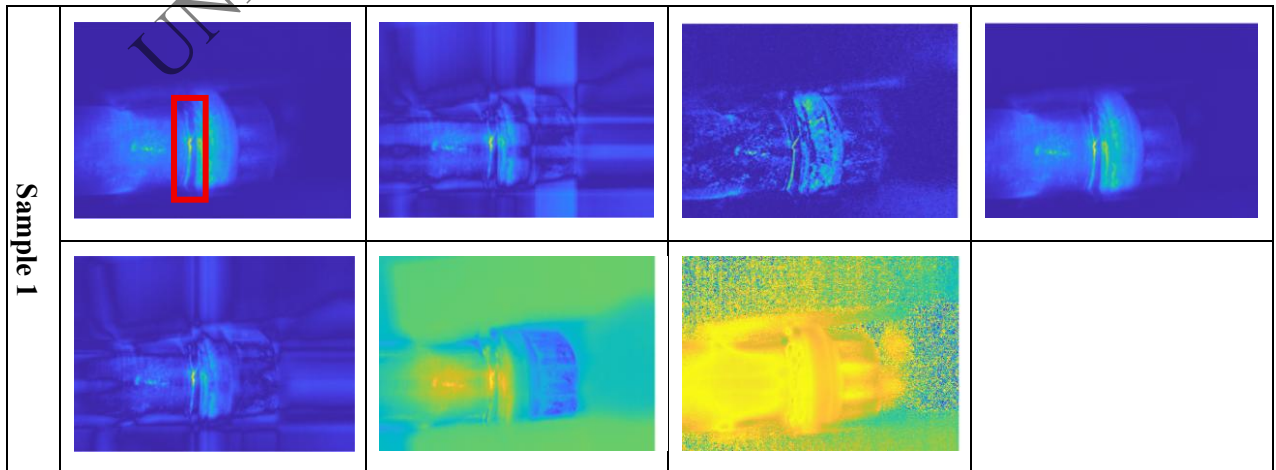
$$AUROC = \int_0^1 TPR(FPR) d(FPR) \quad (16)$$

An AUROC value of 1.0 indicates perfect separation, while 0.5 corresponds to random performance. Since AUROC reflects ranking and SNR reflects absolute contrast, both metrics together provide a complete evaluation. ROI coordinates are fixed for all specimens and methods.

For each method that produces multiple output frames (one per input keyframe or temporal component), the output frame with the highest mean pixel intensity within the fixed defect ROI (S_{area}) was selected as the representative result for both visualization and quantitative metric computation. This selection criterion was applied uniformly to all seven compared methods prior to any SNR or AUROC computation, ensuring an unbiased comparison.

Six methods from classical thermographic analysis to tensor decomposition are reviewed and compared with the proposed KFE-IRCUR. PPT [30] uses phase images obtained through fast Fourier transform (FFT) of the thermal sequence to nullify the influence of non-uniform heating amplitude. TSR [31] fits a low-order polynomial to the log-log thermal decay curve of each pixel with the coefficient deviations used as defect indicators; both form standard classical baselines in ECPT literature. CP-ALS-QR [37] stands for canonical polyadic (CP) tensor decomposition by alternating least squares with QR factorization. OSTD [35] updates the low-rank subspace estimate is updated on a frame-by-frame basis. IRTPCA [36] refers to robust tensor PCA with an improved tensor nuclear norm. LRTSMoG [38] utilizes low-rank tensor decomposition combined with a Mixture-of-Gaussians noise model for non-Gaussian noise management. Computation times are exclusively quoted, excluding data loading, normalization, and metric computation, in all cases.

Figure 4: Visual detection results for all five specimens. Top row (left to right): Proposed, LRTSMoG [38], IRTPCA [36], OSTD [35]. Bottom row (left to right): CP-ALS-QR [37], TSR [31], PPT [30]



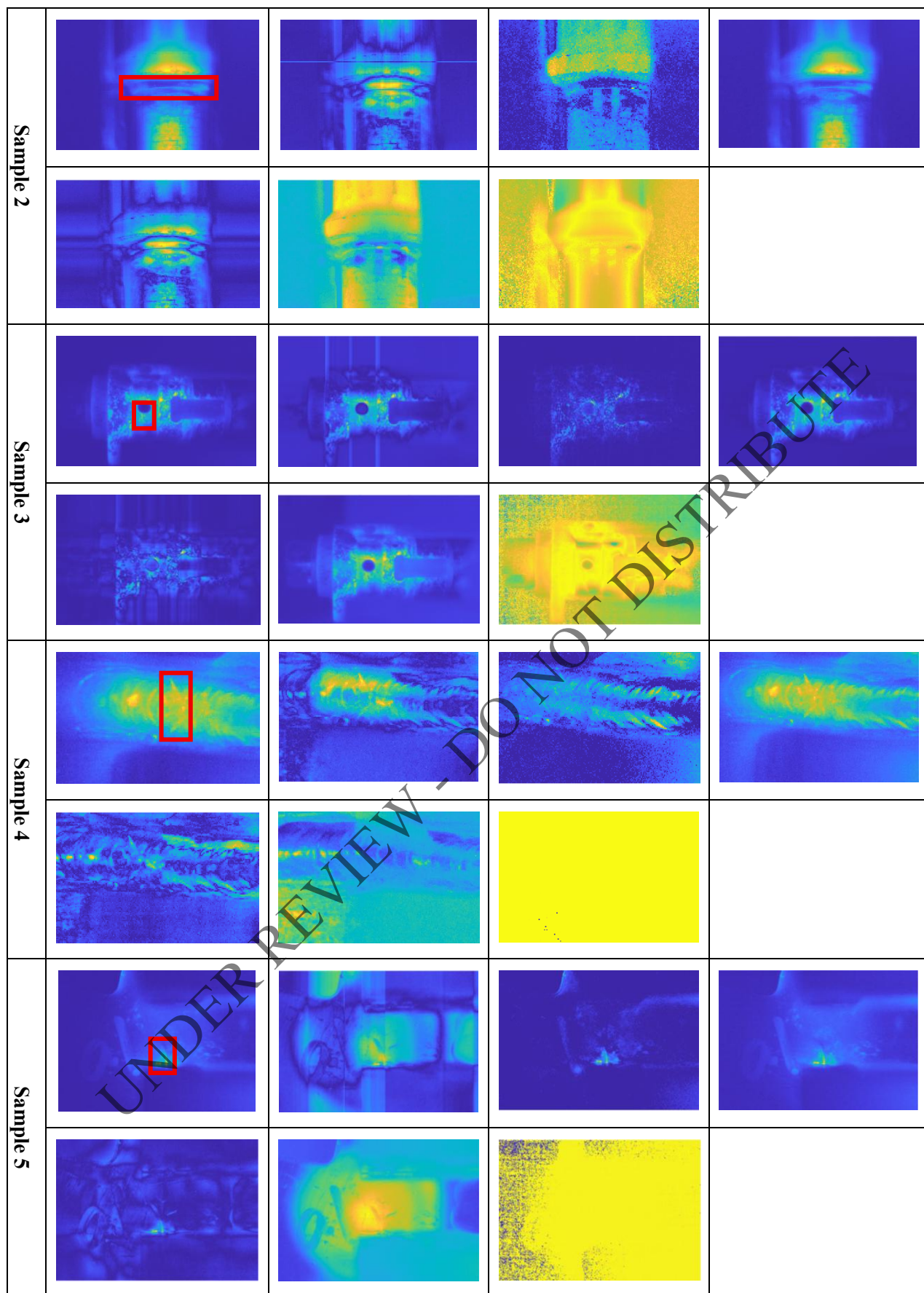


Figure 4 shows the qualitative detection results for all five specimens. Quantitative computation time, SNR, and AUROC values are reported in Tables II, III, and IV respectively; each sample is discussed below across all three performance dimensions.

Samples 1 and 2 are threaded shaft specimens with natural fatigue cracks at the thread root. In addition, the crack is narrow and discontinuous and induced structured electromagnetic heating along the thread profile creates a background thermal pattern, which directly competes with the crack signal. Discrimination of the crack from the thread-root temperature gradient is impeded as PPT returns a low contrast to both specimens (SNR 4.86 dB and 5.28 dB). TSR is 11.61 dB and 29.65 dB, respectively, but crack boundaries are blurred. Notably, CP-ALS-QR yields suboptimal results in the entire dataset on the two specimens (13.41 dB and 19.53 dB), confirming that the narrow fatigue crack cannot be distinguished through CP decompositions from the background, which is driven by the geometry. IRTPCA and LRTSMoG return moderate to low values (15.64 dB and 15.91 dB for IRTPCA, and 3.62 dB and 18.58 dB for LRTSMoG), confirming their struggle to suppress the structured geometry. Meanwhile, OSTD achieves much higher contrast (39.58 dB and 37.84 dB) by better accounting for the non-Gaussian noise structure, though with a visible artifact along the thread profile. The proposed algorithm achieves 43.94 dB and 41.69 dB and its runtime is 0.70 s and 0.41 s compared to 52.37 s and 89.95 s for OSTD and 32.30 s and 103.75 s for IRTPCA.

Sample 3 is a flanged component with a natural surface crack on a geometrically complex body. Non-uniform electromagnetic excitation results in differential heating of various surface locations independent of the crack, which makes background suppression the main challenge. PPT, CP-ALS-QR reach (3.88 and 36.20) dB, respectively while IRTPCA reaches 25.71 dB. LRTSMoG achieves 42.38 dB, while OSTD reaches a higher 48.14 dB but with structured artefacts at component edges. The proposed method achieves 52.75 dB, the highest on this specimen, and well within the strongly separable detection range, with an AUROC of 1.00 and a processing time of 0.72 s versus 103.22 s for OSTD. The keyframe selection step appears to be most beneficial with cooling-phase frames after the geometry-driven heating has started stabilizing, but the anomaly caused by the crack, which takes longer to recede due to the obstruction of the heat-flow path, stays relatively high.

Sample 4 is a weld joint specimen where the weld bead generates a strong spatially structured thermal signature from dense surface scales that can produce spuriously elevated SNR for any methodology that does not explicitly distinguish based on response to weld-bead features versus the crack. PPT returns its only negative value across the dataset at -0.67 dB, and CP-ALS-QR yields 17.48 dB. TSR recovers at 14.37 dB with a faint response. IRTPCA and LRTSMoG achieve partial detection in this sample at 21.61 dB and 24.87 dB. OSTD reaches 33.90 dB but a visual inspection shows its response is partially following the weld bead structure rather than the crack location alone. The proposed method returns 38.88 dB and an AUROC of 1.00, exceeding OSTD in quality of detection, completing in 0.42 s versus 19.55 s.

Sample 5 is a flat cast plate with a natural surface crack next to holes and surface printing that is characterized by inhomogeneous heating and reliable localised thermal patterning unrelated to the defect. PPT reaches only 0.24 dB while CP-ALS-QR yields 27.90 dB. TSR shows a large lead over most other methods with 37.66 dB, its polynomial fitting is well compatible with the simple thermal decay of the flat geometry and successfully suppresses the hole-induced background. LRTSMoG yields 32.08 dB with an AUROC of 0.92, while OSTD reaches 34.53 dB with an AUROC of 1.00. The proposed method returns 38.73 dB with an AUROC rating of 1.00. The proposed method performs above TSR for this specimen, in 0.73 s instead of 1.86 s for TSR.

Table II: Computational time in seconds for all methods across five specimens

Sample	PPT	TSR	CP-ALS	OSTD	IRTPCA	LRTSMoG	Proposed
Sample 1	18.44	3.14	1.37	52.37	32.30	66.71	0.70
Sample 2	50.93	5.15	2.72	89.95	103.75	118.95	0.41
Sample 3	23.33	4.92	1.27	103.22	67.98	84.10	0.72
Sample 4	4.90	2.29	0.49	19.55	19.16	26.07	0.42
Sample 5	2.70	1.86	0.07	6.12	3.93	5.10	0.73
Mean	20.06	3.47	1.19	54.24	45.42	60.19	0.59

Computation times are given in Table II. Further, the proposed method computed an average of 0.59 s across all five specimens, the lowest mean of all the evaluated methods. Two factors compound to produce this saving: key frame extraction reduces the input raw frames down to seven representative frames; and IRCUR circumvents full SVD operations by working on subsampled rows and columns. IRTPCA reaches 103.75 s on Sample 2, precluding practical deployment, and OSTD recorded 54.24 s mean.

Table III: SNR in dB for all seven methods across five specimens

Sample	PPT	TSR	CP-ALS	OSTD	IRTPCA	LRTSMoG	Proposed
Sample 1	4.86	11.61	13.41	39.58	15.64	3.62	43.94
Sample 2	5.28	29.65	19.53	37.84	15.91	18.58	41.69
Sample 3	3.88	44.36	36.20	48.14	25.71	42.38	52.75
Sample 4	-0.67	14.37	17.48	33.90	21.61	24.87	38.88
Sample 5	0.24	37.66	27.90	34.53	32.58	32.08	38.73
Mean	2.72	27.53	22.90	38.80	22.29	24.31	43.20

Table III provides signal to noise ratio (SNR) values. The proposed method has a mean value of 43.20 dB, followed by OSTD at 38.80 dB, and TSR at 27.53 dB. CP-ALS-QR and IRTPCA scores are lower on average (22.90 dB and 22.29 dB, respectively), indicating that background noise dominates, rather than defect response. However, the proposed method leads on all five samples, outperforming all other methods on every single specimen.

Table IV: AUROC for all methods across five specimens

Sample	PPT	TSR	CP-ALS	OSTD	IRTPCA	LRTSMoG	Proposed
Sample 1	0.96	0.99	0.81	0.99	0.81	0.71	1.00
Sample 2	0.95	0.99	0.91	1.00	0.84	0.92	1.00
Sample 3	0.96	1.00	0.98	0.98	0.94	0.91	1.00
Sample 4	0.81	0.87	0.94	1.00	0.96	0.90	1.00
Sample 5	0.75	1.00	0.97	1.00	0.86	0.92	1.00
Mean	0.88	0.97	0.92	1.00	0.88	0.87	1.00

Table IV provides AUROC values. The proposed method and OSTD achieved a mean of 1.00; TSR scored 0.97, CP-ALS-QR scored 0.92, IRTPCA and PPT scored 0.88, and LRTSMoG was the least performing with 0.87. The high AUROC of OSTD against a lower average SNR of 38.80 dB underlines the differences between threshold-independent rank-ordering and absolute contrast, that is a ranking of defect pixels can be guaranteed while absolute separation may not be adequate for real applications. The proposed method achieves both a mean AUROC of 1.00, and the highest mean SNR of 43.20 dB, thereby confirming that its defect signals are not only strongly rank-ordered but also are absolutely well-separated from background.

Across all five specimens, none of the six baselines replicate the performance-efficiency balance achieved by the proposed KFE-IRCUR. While OSTD has detection quality closest to the proposed KFE-IRCUR, it comes at a computational cost of about 92 times more. TSR on the other hand deteriorates significantly (14.37 dB on Sample 4 against 38.88 dB on the proposed method) with respect to more structurally complex specimens despite having an advantage on the comparatively flat cast plate. CP-ALS-QR and IRTPCA return much lower mean SNR values, confirming that reliable detection of structured thermal backgrounds is impossible. OSTD attains a high AUROC, although it is slow (54.24 s mean) and produces artefact-prone images on geometrically complex specimens. No other evaluated algorithm achieves the triad of the lowest mean processing time (0.59 s), highest mean SNR (43.20 dB), and mean AUROC of 1.00 as achieved by the proposed method.

V. Conclusion

Based on the progress presented in this paper, we conclude that our new methodology significantly improves resolution and reduces the computation time for the defect detection in Eddy Current Pulsed Thermography (ECPT). By integrating a keyframe extraction (KFE) mechanism using entropy and density clustering, we have reduced the computational burden associated with processing the entire thermal sequence. This strategic selection of keyframes significantly reduces the processing load without compromising important information about the defects. In addition, the introduction of accelerated CUR decomposition based on low-rank sparse decomposition has proved to be fundamental for efficiently isolating weak defect signals from noisy thermal sequences. This method not only improves the accuracy of defect detection, but also speeds up the computation by utilising a smaller matrix to approximate low-rank sparse components. Experimental results validated on a variety of samples consistently demonstrate that the proposed method outperforms existing state-of-the-art algorithms in terms of the speed and accuracy of extracting the defect information.

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